# Project Phase 2: Dynamic Website with Online Ordering

**Due Date:** End of Week 4 **Points:** 100

## Scenario

The Riverside Cafe's new static website has been a hit! Customers love it, but they immediately started asking for more: the ability to place orders online. To meet this demand, The Riverside Cafe's development team has created a dynamic web application that can take orders and display an order history.

Your role as the cloud engineer-architect is to evolve the cloud infrastructure to support this new application. The simple object storage solution from Phase 1 is no longer sufficient. You need to provision a server environment that can run the application code and connect to a database.

## Project Goals

1. The infrastructure must support a web application which has been built on a standard LAMP stack (Linux, Apache, MySQL/MariaDB, PHP).
2. The application server must be placed within a secure network, and sensitive information like database credentials must not be stored in the application's code.
3. To prepare for future growth and disaster recovery, you need to create a standardized, reusable image of your configured server.

## Technical Requirements

- You will be provided with an application package (`Cafe-app.zip`) and a document with "Installation Notes" from the development team. These notes will contain the necessary commands to install the application and its dependencies on a standard Linux server. **You are not expected to write or debug the application code.**
- Your infrastructure must be defined within a custom virtual network (e.g., AWS VPC, Azure VNet). You should not use the default network provided by the cloud provider.
- All resources must be documented.

## Submission Instructions

You will submit a single PDF document to Moodle. This document must contain all parts of the deliverable outlined below.

**Deliverable Breakdown**

- Provide a public URL to your live, functioning dynamic web application (e.g., `http://<your-server-ip>/Cafe`).

- Include a screenshot of the "Order History" page showing at least two test orders you have placed.
- **Virtual Network (VPC/VNet) Architecture Diagram:** Provide an architecture diagram of your virtual network. It must include the overall VPC/VNet CIDR block, at least one public and one private subnet, an Internet Gateway, and a NAT Gateway. Justify your choice of CIDR blocks and your subnetting strategy. Explain whether you placed the application's VM in the public or private subnet and justify your decision based on security.
- **Security Group Configuration:** Detail the inbound and outbound rules for the security group(s) associated with your VM. For each rule, explain its purpose, adhering to the principle of least privilege.
- **Secrets Management:** Describe the cloud service you used to manage this secret securely and how the VM was given permission to access it.
- **Reusable Server Image and Configuration:** Provide a high-level log of the key steps you took to configure the VM and deploy the application. What is the name of a reusable server image in your chosen cloud provider? Explain two major benefits of creating this image for scalability and disaster recovery. Provide a screenshot showing the created image.

## Grading Rubric

| Criteria | Description | Points |
|---|---|---|
| **Live Application** | URL and screenshot prove a functioning dynamic application. | 10 |
| **Network Architecture** | Diagram is clear, complete, and well-designed. Justifications for network and instance placement are sound. | 30 |
| **Security Configuration** | Security group rules are appropriate, well-explained, and follow the principle of least privilege. Secrets management strategy is sound and correctly implemented. | 30 |
| **Golden Image** | A server image is created and the purpose and benefits of it are clearly explained. | 20 |
| **Professionalism & Clarity** | The submitted document is well-organized and clear. It also clearly discusses the team's working communication and individual contributions for the phase. | 10 |

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| **Total** | | **100** |